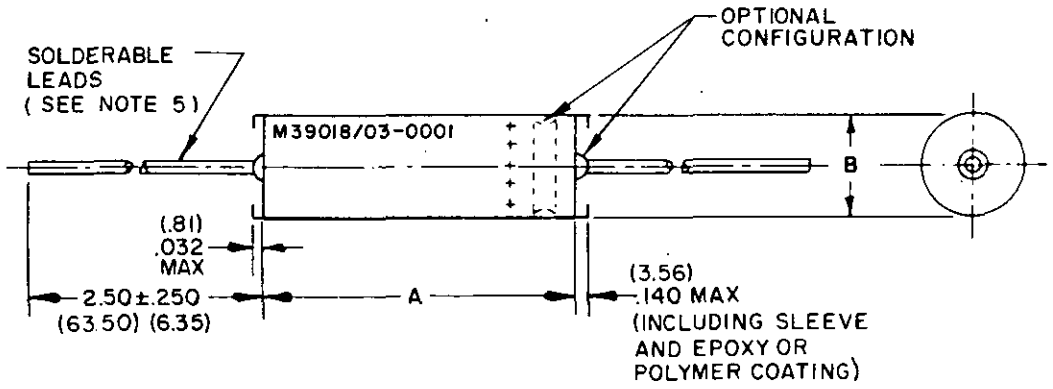
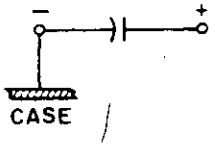
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (ALUMINUM OXIDE) (POLARIZED), CU17 (INSULATED)

This limited coordination Military specification sheet has been prepared by the Department of the Air Force based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification Sheet MIL-C-39018/3A. It is subject to modification. However, pending its promulgation as a coordinated Military specification sheet, it may be used in procurement.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-39018.





Ⓒ denotes changes.

FSC 5910

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Case code	Dimensions	
	A± .031(.79)	B± .015(.38) See note 6
D1	1.125(28.58)	.625(15.88)
D2	1.125(28.58)	.750(19.05)
D3	1.125(28.58)	.875(22.23)
D4	1.625(41.28)	.625(15.88)
D5	1.625(41.28)	.750(19.05)
D6	1.625(41.28)	.875(22.23)
D7	1.625(41.28)	1.000(25.40)
E1	2.125(53.98)	.625(15.88)
E2	2.125(53.98)	.750(19.05)
E3	2.125(53.98)	.875(22.23)
E4	2.125(53.98)	1.000(25.40)
E5	2.625(66.68)	.625(15.88)
E6	2.625(66.68)	.75 (19.05)
E7	2.625(66.68)	.875(22.23)
F1	2.625(66.68)	1.000(25.40)
F2	3.125(79.38)	.625(15.88)
F3	3.125(79.38)	.875(22.23)
F4	3.125(79.38)	1.000(25.40)
F5	3.625(92.08)	.750(19.05)
F6	3.625(92.08)	.875(22.23)
F7	3.625(92.08)	1.000(25.40)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parentheses.
4. These capacitors are not intended to be mounted by their leads.
5. For .625(15.88 mm) diameter case, the leads shall be .032(.81 mm) +.008(.20 mm) -.002(.05 mm) diameter and for all other diameter cases the leads shall be .040 (1.02 mm) +.008(.20 mm) -.002(.05 mm) diameter.
6. The "B" dimension shall be not more than .008 larger than the specified bare case diameter.

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TABLE I. Style CU17 capacitors.

DC rated voltage at 85°C	Capacitance	DC surge voltage 85°C	Dissipation factor		Maximum impedance at -55°C and 120 Hz	DC leakage		Maximum AC ripple current 120 Hz @ 85°C	Case code	Dash number Style CU17 Capacitance tolerance		
			25°C	85°C		25°C	85°C			-10+30	-10+50	-10+75
Volts	µF	Volts	Percent		Ω	µA		mA				
7	680	10	40	50	5	30	183	828	D1	0601	---	0701
7	1,000	10	40	50	9.3	35	212	1,089	D2	0602	---	0702
7	1,500	10	40	50	6.2	43	259	1,470	D4	0603	---	0703
7	2,200	10	40	50	4.2	50	300	2,040	E1	0604	---	0704
7	3,300	10	40	50	2.9	66	396	2,608	D6	0605	---	0705
7	4,700	10	40	50	2.0	75	450	3,313	D7	0606	---	0706
7	6,800	10	40	50	1.3	90	540	5,164	F5	0607	---	0707
7	10,000	10	40	50	.93	111	670	6,723	F4	0608	---	0708
7	12,000	10	40	50	.77	127	764	7,929	F7	0609	---	0709
10	470	15	35	40	8.5	35	212	379	D1	0610	---	0710
10	1,000	15	35	40	4.0	50	300	1,320	D4	0611	---	0711
10	1,500	15	35	40	2.7	61	367	1,851	E1	0612	---	0712
10	2,200	15	35	40	1.8	70	424	2,320	D6	0613	---	0713
10	3,300	15	35	40	1.2	93	558	3,253	E3	0614	---	0714
10	4,700	15	35	40	.85	110	663	4,681	F5	0615	---	0715
10	6,800	15	35	40	.59	127	764	6,106	F6	0616	---	0716
10	8,200	15	35	40	.50	150	900	7,140	F7	0617	---	0717
15	390	20	25	30	10	38	232	766	D1	0618	---	0718
15	680	20	25	30	5.9	49	296	1,108	D2	0619	---	0719
15	1,000	20	25	30	4.0	61	367	1,689	E1	0620	---	0720
15	1,500	20	25	30	2.7	75	450	2,266	E2	0621	---	0721
15	2,200	20	25	30	1.8	86	519	3,051	E6	0622	---	0722
15	3,300	20	25	30	1.2	114	687	3,883	E4	0623	---	0723
15	4,700	20	25	30	0.8	136	821	5,676	F6	0624	---	0724
15	6,800	20	25	30	0.60	156	936	7,305	F7	0625	---	0725
30	150	40	25	30	27	33	201	474	D1	0626	---	0726
30	220	40	25	30	14	47	284	630	D2	0627	---	0727
30	330	40	25	30	12	51	307	846	D4	0628	---	0728
30	470	40	25	30	10	54	328	996	D3	0629	---	0729
30	680	40	25	30	6.0	69	418	1,546	E5	0630	---	0730
30	1,000	40	25	30	4.0	86	519	1,998	E3	0631	---	0731
30	1,500	40	25	30	2.7	110	663	2,724	E7	0632	---	0732
30	2,200	40	25	30	1.8	122	734	3,876	F6	0633	---	0733
30	2,700	40	25	30	1.5	136	821	4,261	F4	0634	---	0734

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TABLE I. Style CU17 capacitors. - Continued

DC rated voltage at 85°C	Capacitance	DC surge voltage 85°C	Dissipation factor		Maximum impedance at -55°C and 120 Hz	DC leakage		Maximum AC ripple current 120 Hz @ 85°C	Case code	Dash number Style CU17 Capacitance tolerance		
			25°C	85°C		25°C	85°C			-10+30	-10+50	-10+75
Volts	μF	Volts	Percent		Ω	μA		mA				
50	100	60	10	30	26	37	222	387	D1	0635	---	0735
50	150	60	10	30	15	46	276	519	D2	0636	---	0736
50	220	60	10	30	13	55	335	679	D3	0637	---	0737
50	330	60	10	30	8.1	66	396	928	D5	0638	---	0738
50	470	60	10	30	5.6	79	474	1,267	E2	0639	---	0739
50	680	60	10	30	3.9	90	540	1,540	D7	0640	---	0740
50	1,000	60	10	30	2.7	111	670	2,136	E4	0641	---	0741
50	1,500	60	10	30	1.8	147	883	3,169	F4	0642	---	0742
50	1,800	60	10	30	1.5	196	1,196	3,545	F7	0699	---	0799
75	47	90	10	30	57	29	174	264	D1	0643	---	0743
75	68	90	10	30	39	37	225	348	D2	0644	---	0744
75	100	90	10	30	27	43	259	459	D3	0645	---	0745
75	150	90	10	30	18	53	318	625	D5	0646	---	0746
75	220	90	10	30	13	68	410	867	E2	0647	---	0747
75	330	90	10	30	8.1	80	483	1,276	E7	0648	---	0748
75	470	90	10	30	5.7	91	551	1,659	F3	0649	---	0749
75	680	90	10	30	3.9	110	662	2,134	F4	0650	---	0750
75	820	90	10	30	3.2	122	734	2,524	F7	0651	---	0751
100	22	125	10	30	125	25	150	177	D1	0652	0752	---
100	47	125	10	30	57	33	201	288	D2	0653	0753	---
100	68	125	10	30	39	40	241	375	D3	0654	0754	---
100	100	125	10	30	27	50	300	646	F2	0655	0755	---
100	150	125	10	30	18	61	367	721	D7	0656	0756	---
100	220	125	10	30	13	79	474	1,000	E4	0657	0757	---
100	330	125	10	30	8.1	93	563	1,498	F6	0658	0758	---
100	470	125	10	30	5.6	171	670	1,909	F7	0659	0759	---
150	18	175	10	30	150	25	155	156	D1	0660	0760	---
150	33	175	10	30	81	36	217	252	D4	0661	0761	---
150	47	175	10	30	57	41	246	300	D3	0662	0762	---
150	68	175	10	30	39	49	296	463	E5	0663	0763	---
150	100	175	10	30	27	61	367	559	D7	0664	0764	---
150	150	175	10	30	18	77	464	817	E7	0665	0765	---
150	220	175	10	30	12	86	519	1,057	F1	0666	0766	---
150	330	175	10	30	8.0	114	687	1,521	F7	0667	0767	---

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TABLE I. Style CU17 capacitors. - Continued

DC rated voltage at 85°C	Capacitance	DC surge voltage 85°C	Dissipation factor		Maximum impedance at -55°C and 120 Hz	DC leakage		Maximum AC ripple current 120 Hz @ 85°C	Case code	Dash number Style CU17 Capacitance tolerance		
			25°C	85°C		25°C	85°C			-10+30	-10+50	-10+75
Volts	μF	Volts	Percent		Ω	μA		mA				
200	15	225	10	30	180	27	164	139	D1	0668	0768	---
200	33	225	10	30	81	41	244	246	D3	0669	0769	---
200	47	225	10	30	57	47	284	337	E1	0670	0770	---
200	68	225	10	30	39	59	354	453	E5	0671	0771	---
200	100	225	10	30	27	70	424	549	D7	0672	0772	---
200	150	225	10	30	18	86	519	769	E4	0673	0773	---
200	220	225	10	30	12	109	657	1,129	F4	0674	0774	---
200	270	225	10	30	9.8	120	722	1,345	F7	0675	0775	---
250	10	275	10	30	250	25	150	115	D1	0676	0776	---
250	20	275	10	30	125	35	210	205	D4	0677	0777	---
250	33	275	10	30	80	45	270	342	E5	0678	0778	---
250	39	275	10	30	65	50	300	361	F2	0679	0779	---
250	68	275	10	30	40	67	402	438	D7	0680	0780	---
250	82	275	10	30	35	80	480	546	E4	0681	0781	---
250	120	275	10	30	25	94	564	813	F6	0682	0782	---
250	180	275	10	30	15	111	666	1,063	F7	0683	0783	---
300	8.2	325	15	30	320	25	150	94	D1	0684	0784	---
300	15	325	15	30	170	35	210	159	D4	0685	0785	---
300	27	325	15	30	93	48	288	238	D5	0686	0786	---
300	47	325	15	30	54	63	378	397	E6	0687	0787	---
300	68	325	15	30	37	77	462	522	E7	0688	0788	---
300	82	325	15	30	31	87	522	622	F3	0689	0789	---
300	120	325	15	30	21	96	576	813	F6	0690	0790	---
300	150	325	15	30	17	112	672	970	F7	0691	0791	---
350	5.6	375	15	30	450	25	150	73	D1	0692	0792	---
350	20	375	15	30	125	42	252	213	E1	0693	0793	---
350	27	375	15	30	93	53	318	300	F2	0694	0794	---
350	47	375	15	30	54	66	396	361	D7	0695	0795	---
350	56	375	15	30	45	75	450	465	E7	0696	0796	---
350	82	375	15	30	31	86	516	657	F3	0697	0797	---
350	120	375	15	30	21	112	672	868	F7	0698	0798	---

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TABLE II. Substitutability data.

Dash no. in MIL-C-0039018/3B and C (USAF)			Substitutable for dash no. in								
			MIL-C-39018/3A			MIL-C-39018/3					
Style CU17			Style CU16			Style CU16			Style CU17		
Capacitance tolerance			Capacitance tolerance			Capacitance tolerance			Capacitance tolerance		
-10+30	-10+50	-10+75	-10+30	-10+50	-10+75	-10+30	-10+50	-10+75	-10+30	-10+50	-10+75
0601	---	0701	0401	---	0501	0001	---	0002	0003	---	0004
0602	---	0702	0402	---	0502	0005	---	0006	0007	---	0008
0603	---	0703	0403	---	0503	0009	---	0010	0011	---	0012
0604	---	0704	0404	---	0504	0013	---	0014	0015	---	0016
0605	---	0705	0405	---	0505	0017	---	0018	0019	---	0020
0606	---	0706	0406	---	0506	0021	---	0022	0023	---	0024
0607	---	0707	0407	---	0507	0025	---	0026	0027	---	0028
0608	---	0708	0408	---	0508	0029	---	0030	0031	---	0032
0609	---	0709	0409	---	0509	0033	---	0034	0035	---	0036
0610	---	0710	0410	---	0510	0037	---	0038	0039	---	0040
0611	---	0711	0411	---	0511	0041	---	0042	0043	---	0044
0612	---	0712	0412	---	0512	0045	---	0046	0047	---	0048
0613	---	0713	0413	---	0513	0049	---	0050	0051	---	0052
0614	---	0714	0414	---	0514	0053	---	0054	0055	---	0056
0615	---	0715	0415	---	0515	0057	---	0058	0059	---	0060
0616	---	0716	0416	---	0516	0061	---	0062	0063	---	0064
0617	---	0717	0417	---	0517	0065	---	0066	0067	---	0068
0618	---	0718	0418	---	0518	0069	---	0070	0071	---	0072
0619	---	0719	0419	---	0519	0073	---	0074	0075	---	0076
0620	---	0720	0420	---	0520	0077	---	0078	0079	---	0080
0621	---	0721	0421	---	0521	0081	---	0082	0083	---	0084
0622	---	0722	0422	---	0522	0085	---	0086	0087	---	0088
0623	---	0723	0423	---	0523	0089	---	0090	0091	---	0092
0624	---	0724	0424	---	0524	0093	---	0094	0095	---	0096
0625	---	0725	0425	---	0525	0097	---	0098	0099	---	0100
0626	---	0726	0426	---	0526	0101	---	0102	0103	---	0104
0627	---	0727	0427	---	0527	0105	---	0106	0107	---	0108
0628	---	0728	0428	---	0528	0109	---	0110	0111	---	0112
0629	---	0729	0429	---	0529	0113	---	0114	0115	---	0116
0630	---	0730	0430	---	0530	0117	---	0118	0119	---	0120
0631	---	0731	0431	---	0531	0121	---	0122	0123	---	0124
0632	---	0732	0432	---	0532	0125	---	0126	0127	---	0128
0633	---	0733	0433	---	0533	0129	---	0130	0131	---	0132
0634	---	0734	0434	---	0534	0133	---	0134	0135	---	0136
0635	---	0735	0435	---	0535	0137	---	0138	0139	---	0140
0636	---	0736	0436	---	0536	0141	---	0142	0143	---	0144
0637	---	0737	0437	---	0537	0145	---	0146	0147	---	0148
0638	---	0738	0438	---	0538	0149	---	0150	0151	---	0152

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TABLE II. Substitutability data. - Continued

Dash no. in MIL-C-0039018/3B and C (USAF)			Substitutable for dash no. in								
			MIL-C-39018/3A			MIL-C-39018/3					
Style CU17			Style CU16			Style CU16			Style CU17		
Capacitance tolerance			Capacitance tolerance			Capacitance tolerance			Capacitance tolerance		
-10+30	-10+50	-10+75	-10+30	-10+50	-10+75	-10+30	-10+50	-10+75	-10+30	-10+50	-10+75
0639	---	0739	0439	---	0539	0153	---	0154	0155	---	0156
0640	---	0740	0440	---	0540	0157	---	0158	0159	---	0160
0641	---	0741	0441	---	0541	0161	---	0162	0163	---	0164
0642	---	0742	0442	---	0542	0165	---	0166	0167	---	0168
0643	---	0743	0443	---	0543	0169	---	0170	0171	---	0172
0644	---	0744	0444	---	0544	0173	---	0174	0175	---	0176
0645	---	0745	0445	---	0545	0177	---	0178	0179	---	0180
0646	---	0746	0446	---	0546	0181	---	0182	0183	---	0184
0647	---	0747	0447	---	0547	0185	---	0186	0187	---	0188
0648	---	0748	0448	---	0548	0189	---	0190	0191	---	0192
0649	---	0749	0449	---	0549	0193	---	0194	0195	---	0196
0650	---	0750	0450	---	0550	0197	---	0198	0199	---	0200
0651	---	0751	0451	---	0551	0201	---	0202	0203	---	0204
0652	0752	---	0452	0552	---	0205	0206	---	0207	0208	---
0653	0753	---	0453	0553	---	0209	0210	---	0211	0212	---
0654	0754	---	0454	0554	---	0213	0214	---	0215	0216	---
0655	0755	---	0455	0555	---	0217	0218	---	0219	0220	---
0656	0756	---	0456	0556	---	0221	0222	---	0223	0224	---
0657	0757	---	0457	0557	---	0225	0226	---	0227	0228	---
0658	0758	---	0458	0558	---	0229	0230	---	0231	0232	---
0659	0759	---	0459	0559	---	0233	0234	---	0235	0236	---
0660	0760	---	0460	0560	---	0237	0238	---	0239	0240	---
0661	0761	---	0461	0561	---	0241	0242	---	0243	0244	---
0662	0762	---	0462	0562	---	0245	0246	---	0247	0248	---
0663	0763	---	0463	0563	---	0249	0250	---	0251	0252	---
0664	0764	---	0464	0564	---	0253	0254	---	0255	0256	---
0665	0765	---	0465	0565	---	0257	0258	---	0259	0260	---
0666	0766	---	0466	0566	---	0261	0262	---	0263	0264	---
0667	0767	---	0467	0567	---	0265	0266	---	0267	0268	---
0668	0768	---	0468	0568	---	0269	0270	---	0271	0272	---
0669	0769	---	0469	0569	---	0273	0274	---	0275	0276	---
0670	0770	---	0470	0570	---	0277	0278	---	0279	0280	---
0671	0771	---	0471	0571	---	0281	0282	---	0283	0284	---
0672	0772	---	0472	0572	---	0285	0286	---	0287	0288	---
0673	0773	---	0473	0573	---	0289	0290	---	0291	0292	---
0674	0774	---	0474	0574	---	0293	0294	---	0295	0296	---
0675	0775	---	0475	0575	---	0297	0298	---	0299	0300	---

NOTE: Capacitors tested to the high frequency vibration specified herein are suitable replacements for like items tested to low frequency vibration.

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REQUIREMENTS:

Dimensions and configuration:
See figure.

Case: Type - Tubular.
Material - Metal.
Leads - See figure.

DC rated voltage: See table I.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Operating temperature:
-55°C to +85°C.

DC leakage: See table I.

Dissipation factor: See table I.

Low temperature exposure:
DCL - See table I.
ΔC - See table I.
DF - See table I.

Stability at reduced and high temperature:
Step 1 (at +25°C):
DCL - See table I.
ΔC - See table I.
DF - See table I.
Step 2 (at -55°C):
Impedance - See table I.
ΔC - Within the following percentages of the step 1 value:
7 WVDC: 35 percent.
10 to 30 WVDC inclusive: 25 percent.
50 to 350 WVDC inclusive: 20 percent.
Step 3 (at +25°C):
DCL - See table I.
ΔC - Within 5 percent of step 1 value.
DF - See table I.
Step 4 (at +85°C):
DCL - See table I.
ΔC - Within 30 percent of step 1 value.
DF - See table I.
Step 5 (at +25°C):
DCL - See table I.
ΔC - Within 5 percent of step 1 value.
DF - See table I.

Life: Method 108 of MIL-STD-202.
For qualification: Condition F (2,000 hours) at +85°C.
DCL - See table I.
ΔC - Within 10 percent of initial measured value.
DF - Not more than 130 percent of initial requirement.

Life: (Cont)
For quality conformance:
Performance check, condition B (250 hours) at 85°C.
DCL - See table I.
ΔC - Within 10 percent of initial measured value.
DF - See table I.
For quality conformance:
Continuation test (1,750 hours) at 85°C.
DCL - See table I.
ΔC - Within 15 percent of initial measured value.
DF - Not more than 130 percent of initial requirement.

Vibration:
Low frequency - Method 201, MIL-STD-202.
High frequency (when specified, see ordering data) - In accordance with MIL-C-39018 with the following exceptions: Test condition letter A; duration and direction of motion - In accordance with method 204 of MIL-STD-202.

Temperature cycling and immersion:
Temperature cycling: Method 102 of MIL-STD-202, condition D.
Immersion: Method 104 of MIL-STD-202, condition B (2 cycles, 30 minutes).
DCL - See table I.
ΔC - Within 5 percent of initial measured value.
DF - See table I.
Insulation resistance of sleeve - Not less than 100 megohms.

Surge voltage:
DCL - See table I.
ΔC - Within 6 percent of initial measured value.
DF - See table I.

Moisture resistance:
Method 106 of MIL-STD-202.
DCL - See table I.
ΔC - Within 6 percent of initial measured value.
DF - See table I.
Insulation resistance of insulating sleeve - Not less than 100 megohms.

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High temperature exposure: (500 hours at 85°C)
DCL - Not more than 200 percent of initial requirement.
ΔC - Within 10 percent of initial measured value.
DF - Not more than 115 percent of initial requirement.

Reverse voltage:
After 125 hours at 85°C with $1\ ^{+0}_{-.1}$ reverse volt applied to units rated at 7 Vdc and $3\ ^{+0}_{-.3}$ reverse volts applied to all others.

After additional 125 hours at 85°C with rated forward voltage applied:
DCL - Within 125 percent of initial requirement.
ΔC - Within the following percentages of initial measured value:
7 to 30 WVDC inclusive:
+10, -25 percent.
50 to 350 WVDC inclusive:
±10 percent.
DF - See table I.

Marking: In accordance with MIL-C-39018.
Part number:
For low frequency vibration:
M39018/03-(dash number from table I).
For high frequency vibration (see ordering data):
M39018/03-(dash number from table I) followed by letter "H".

Example:

M39018/03-0601H

Ordering data: In addition to the data specified in MIL-C-39018, procurement documents should specify those capacitors which are to be tested to high frequency vibration.

Preparing activity:
Air Force - 11

Agent:
DSA - ES

(Project 5910-F250)

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